

Shun Zhang

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Research Interests Reinforcement learning; foundation models; reasoning and self-improving agents; value alignment.

Experience

- Aug 2025 – Present **Senior Generative AI Engineer**, *NVIDIA*, Santa Clara, CA
Developing simulation-based benchmarks for robot **policy models**.
Post-training and evaluating **vision-language models (VLMs)** for embodied AI capabilities.
- Jun 2024 – Jan 2025 **Founding Member of Technical Staff**, *Asari AI*, San Francisco, CA
Developed a **code agent** that plans, verifies, and discovers new skills and knowledge.
- Jun 2022 – Jun 2024 **Research Scientist**, *MIT-IBM Watson AI Lab*
Research on **reinforcement learning** and **post-training of language models**, with a focus on code generation and reinforcement learning from human feedback (RLHF).
- Aug 2020 – Jun 2022 **Postdoctoral Researcher**, *IBM Research*
Research on **meta-reinforcement learning** and **AI for scientific discovery**.

Education

- Sep 2015 – Apr 2020 **Ph.D. in Computer Science and Engineering**, *University of Michigan*, Ann Arbor, MI
Advisors: Prof. Satinder Singh and Prof. Ed Durfee.
Research on value alignment and AI safety in reinforcement learning.
- Aug 2015 **B.S. and M.S. in Computer Science**, *University of Texas at Austin*, Austin, TX
Undergraduate/master research advisors: Prof. Peter Stone and Prof. Dana Ballard.

Publications and Preprints

- 2026 **Cosmos 3: Omnimodal World Models for Physical AI**
NVIDIA
arXiv 
- 2024 **Improving Reinforcement Learning from Human Feedback with Efficient Reward Model Ensemble**
Shun Zhang, Zhenfang Chen, Sunli Chen, Yikang Shen, Zhiqing Sun, Chuang Gan
arXiv 
- 2024 **Graph-Transformer-based Surrogate Model for Accelerated Converter Circuit Topology Design**
Shaoze Fan, Haoshu Lu, **Shun Zhang**, Ningyuan Cao, Xin Zhang, Jing Li
Design Automation Conference (DAC) 
- 2023 **Adaptive Online Replanning with Diffusion Models**
Siyuan Zhou, Yilun Du, **Shun Zhang**, Mengdi Xu, Yikang Shen, Wei Xiao, Dit-Yan Yeung, Chuang Gan
Conference on Neural Information Processing Systems (NeurIPS) 
- 2023 **Planning with Large Language Models for Code Generation**
Shun Zhang, Zhenfang Chen, Yikang Shen, Mingyu Ding, Joshua B. Tenenbaum, Chuang Gan
International Conference on Learning Representations (ICLR) 
- 2023 **Hyper-Decision Transformer for Efficient Online Policy Adaptation**
Mengdi Xu, Yuchen Lu, Yikang Shen, **Shun Zhang**, Ding Zhao, Chuang Gan
International Conference on Learning Representations (ICLR) 

- 2022 **Prompting Decision Transformer for Few-shot Policy Generalization**
Mengdi Xu, Yikang Shen, **Shun Zhang**, Yuchen Lu, Ding Zhao, Joshua B. Tenenbaum, Chuang Gan
International Conference on Machine Learning (ICML) 
- 2022 **Power Converter Circuit Design Automation using Parallel Monte Carlo Tree Search**
Shaoze Fan, **Shun Zhang**, Jianbo Liu, Ningyuan Cao, Xiaoxiao Guo, Jing Li, Xin Zhang
ACM Transactions on Design Automation of Electronic Systems (TODAES) 
- 2021 **From Specification to Topology: Automatic Power Converter Design via Reinforcement Learning**
Shaoze Fan, Ningyuan Cao, **Shun Zhang**, Jing Li, Xiaoxiao Guo, Xin Zhang
International Conference on Computer Aided Design (ICCAD) 
- 2020 **Efficiently Finding Approximately-Optimal Queries for Improving Policies and Guaranteeing Safety**
Shun Zhang
Ph.D. Dissertation 
- 2020 **Querying to Find a Safe Policy Under Uncertain Safety Constraints in Markov Decision Processes**
Shun Zhang, Edmund H. Durfee, Satinder Singh
AAAI Conference on Artificial Intelligence (AAAI) 
- 2018 **Minimax-Regret Querying on Side Effects for Safe Optimality in Factored Markov Decision Processes**
Shun Zhang, Edmund H. Durfee, Satinder Singh
International Joint Conference on Artificial Intelligence (IJCAI) 
- 2018 **Modeling Sensory-Motor Decisions in Natural Behavior**
Ruohan Zhang, **Shun Zhang**, Matthew H. Tong, Yuchen Cui, Constatin A. Rothkopf, Dana H. Ballard, Mary M. Hayhoe
PLoS Computational Biology 
- 2017 **Approximately-Optimal Queries for Planning in Reward-Uncertain Markov Decision Processes**
Shun Zhang, Edmund H. Durfee, Satinder Singh
International Conference on Automated Planning and Scheduling (ICAPS) 
- 2015 **Determining Placements of Influencing Agents in a Flock**
Katie Genter, **Shun Zhang**, Peter Stone
Autonomous Agents and Multiagent Systems (AAMAS) 
- 2015 **Autonomous Intersection Management for Semi-Autonomous Vehicles**
Tsz-Chiu Au, **Shun Zhang**, Peter Stone
Handbook of Transportation 

Academic Services

Conference Reviewer AAAI 2019, AISTATS 2023-24, CVPR 2023, ICML 2023-24, NeurIPS 2023-25, ICLR 2024-25.